

PHILOSOPHY OF SCIENCE

LESSON 8

KUHN AND POPPER ON SCIENTIFIC PROGRESS - PART 2

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AGENDA: FIRST HALF OF LESSON

Kuhn and Popper on Scientific Progress - i) When is a theory 'true'? Understanding the nature of science and scientific methods ii) How does science develop: Comparing Kuhn's idea of revolutions and Popper's falsification	Kuhn 1970 (section 4 p. 191-198 & section 6 p. 205-207) PoS 5 ABH 2018 ch 3 Popper 1962 (text segments)
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REVISITING KUHN: HOW SCIENCE DEVELOPS II

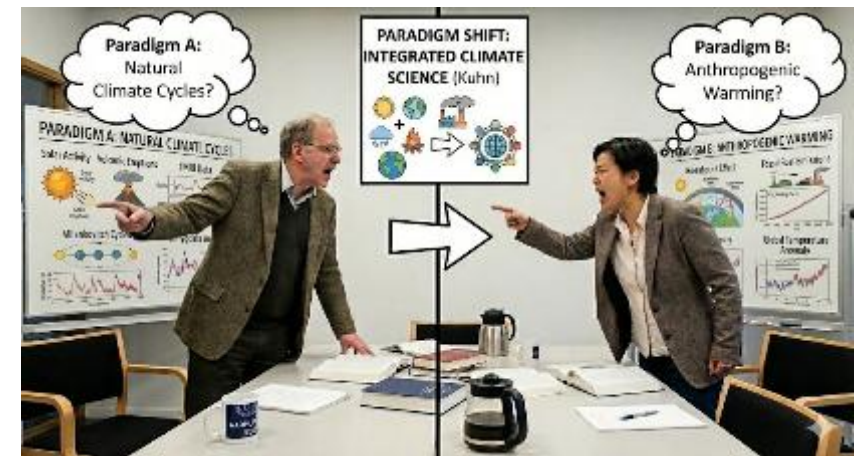


Positivism, pre-Kuhn: “...positivists believed that disputes between rival scientific theories could be solved in a perfectly objective way, comparing the theories directly to the observations...” (PoS 5, p.80)

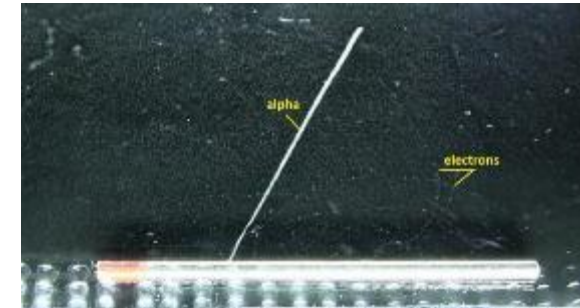
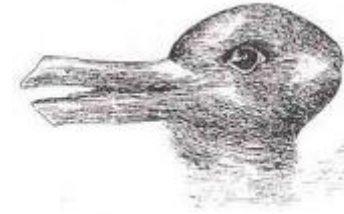
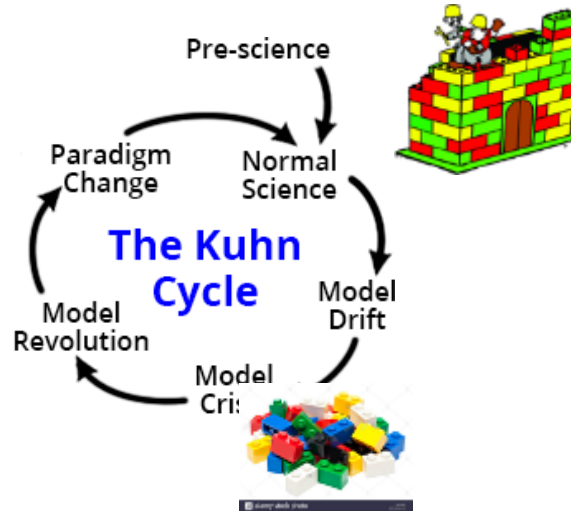
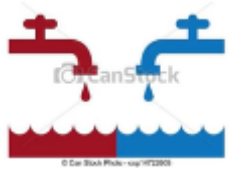
General (before Kuhn / Popper) idea about science and scientists:

- search for truth
- uncover how the world really looks like
- are rational

Implication: better explanations / data should be accepted



REVISITING KUHN: HOW SCIENCE DEVELOPS II



Changing opinion among economists:

Do minimum wages substantially lower employment among low-wage workers?

- 1978 AEA Member Survey: 90% agreed
- 1992 AEA Member Survey: 72% agreed
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Analysis of petition signers (O'Neill 2014):

Labor economists, recent PhDs more likely to support raising minimum wages

Kuhn p. 206: "There is, I think, no theory-independent way to reconstruct phrases like 'really there'; the notion of a match between the ontology of a theory and its "real" counterpart in nature now seems to me illusive in principle" (critique of positivism)



TWO DIFFERENT TYPES OF RESEARCHERS

- **Truth-searchers (Positivists)**

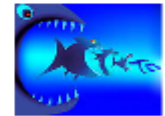
- go where the truth takes them
- external standards for objectivity

- **Problem-solvers (Kuhn)**

- no theory-independent truth to find
- paradigm driven standards for 'objectivity'
- solve problems – that one considers important (who decides?)
- consensus and power dynamics important

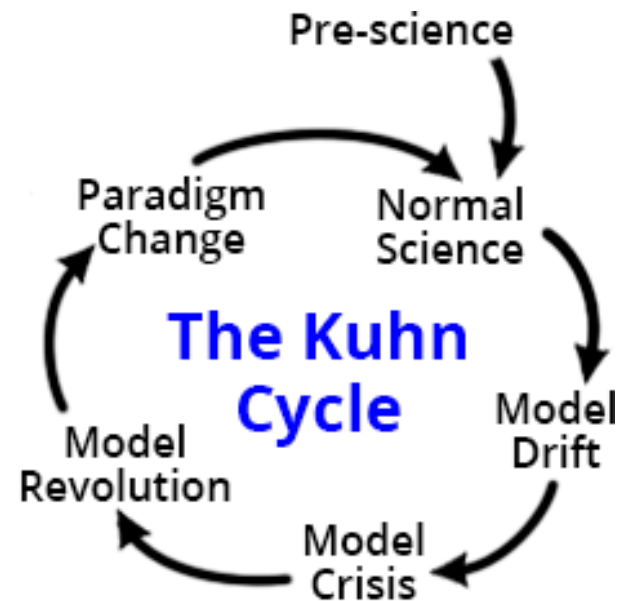
COMPARING DIFFERENT PARADIGMS → SURVIVAL OF THE FITTEST

- Not only about funerals – newer paradigms can be better / not simple relativism
- Yet not better representation of reality, cf. Kuhn 1970: p. 206
→ survival of the fittest, across paradigms
- Criteria for comparison (Kuhn 1970: p. 206)
 - Accuracy of prediction
 - Number of different problems solved
 - Simplicity, scope, compatibility with other disciplines



→ Science revealed to be inherently shaped by institutions and incentives

CONSIDER THIS: IS IT A GOOD THING IF 100% OF SCIENTISTS AGREE ON SOMETHING? (REFLECT 1-2 MIN)



FROM KUHN TO POPPER

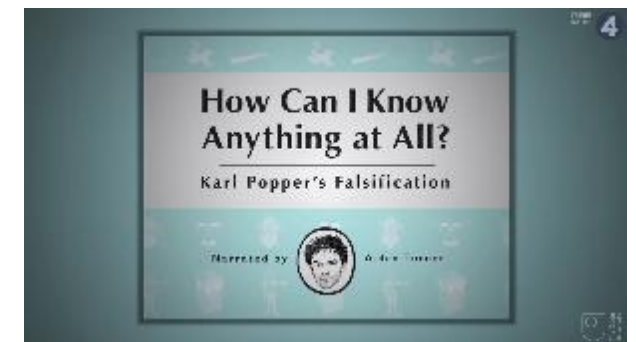
- **Kuhn: History of science** (what do people do)
 - How science **has worked, historically**
- **Popper: Analytical** (what should people do)
 - What is science and how it **should work normatively**
- **Popper's main issues**
 - Problem of induction -> science is deductive
 - Don't go for truth (verification) -> falsify
 - Asymmetry thesis -> one is enough to falsify
 - Realism and tentative truths (-> post-positivism, more in lesson 9)

THE (POSITIVISM) POSITION THAT KUHN & POPPER ARGUE AGAINST



Positivism: Classic 'textbook' conception of science pre-Kuhn (1970)

- "...positivists believed that disputes between rival scientific theories **could be solved in a perfectly objective way**, comparing the theories directly to the observations..." (PoS 5, p.80)
 - The world can be observed as it is / there are neutral facts (PoS p. 80)
 - Theory corresponds to / mirrors world (PoS p. 78) →
 - We can tell if a theory is true / false
 - Theory development as a **rational process** (PoS p. 78-79)



POPPER'S TWO PROBLEMS WITH INDUCTION I

(PURE INDUCTION DOES NOT EXIST)

Idea of pure induction: That we can observe the world and then theorize

“The belief that science proceeds from observation to theory is still so widely and so firmly held that my denial of it is often met with incredulity. I have even been suspected of being insincere--of denying what nobody in his senses can doubt.

Twenty-five years ago I tried to bring home the same point to a group of physics students in Vienna by beginning a lecture with the following instructions: 'Take pencil and paper; carefully observe, and write down what you have observed!' They asked, of course, *what* I wanted them to observe. Clearly the instruction, 'Observe!' is absurd. 13 (It is not even idiomatic, unless the object of the transitive verb can be taken as understood.) Observation is always selective. It needs a chosen object, a definite task, an interest, a point of view, a problem. And its description presupposes a descriptive language, with property words; it presupposes similarity and classification, which in its turn presupposes interests, points of view, and problems.”

Conjectures and Refutations 1962, p. 46

POPPERS TWO PROBLEMS WITH INDUCTION II

THE NEXT SWAN MIGHT BE BLACK

provide us with certain knowledge. In philosophy of science, this is what is called *the problem of induction*, and it has been debated since the eighteenth century (Popper 1974: 42–46).

! **The Problem of Induction:** Even if all our observations of the world have so far yielded the same result, we can never be certain that our next observation will not produce a different result.

ABH: p. 47

! **The Asymmetry Thesis:** Numerous observations can never prove a scientific theory but one observation is enough to disprove it.

ABH: p. 48



Popper sets a high bar for scientific knowledge!

WE PROPOSE A THEORY AND TEST IT I

Yet: We should not try to verify a theory. Too easy to find examples that (seem to) support.

“The most characteristic element in this situation seemed to me the incessant stream of confirmations, of observations which 'verified' the theories in question; and this point was constantly emphasized by their adherents. A Marxist could not open a newspaper without finding on every page confirming evidence for his interpretation of history; not only in the news, but also in its presentation--which revealed the class bias of the paper--and especially of course in what the paper did *not* say. The Freudian analysts emphasized that their theories were constantly verified by their

Conjectures and Refutations 1962, p. 35

WE PROPOSE A THEORY AND TEST IT SEVERELY – TO FALSIFY IT II

Make conjectures (informed guesses) – and then seek to refute/falsify it (!)

! **Falsificationism:** The opposite of verificationism, this is the view that the sciences do not seek to prove but rather to disprove their theories; i.e., the sciences consist of critically testing hypotheses.

ABH: p. 48

“The way in which knowledge progresses, and especially our scientific knowledge, is by unjustified (and unjustifiable) anticipations, by guesses, by tentative solutions to our problems, by *conjectures*. These conjectures are controlled by criticism; that is, by attempted *refutations*, which include severely critical tests. They may survive these tests; but they can never be positively justified: they can neither be established as certainly true nor even as 'probable' (in the sense of the probability calculus). Criticism of our conjectures is of decisive importance: by bringing out our mistakes it

Conjectures and Refutations 1962, p. vii

WHY ONE (OR MANY) TESTS DON'T VERIFY A THEORY

Imagine a theory: Higher financial incentives increase performance in simple tasks

One can test this theory in a wide range of ways

- Different research designs
- Different populations
- Different settings / tasks
 - Increase incentives a lot / a bit
 - Use different kinds of tasks
 - Measure over short/long periods of time
 -

No matter how many tests one has run...one can always think of additional – potentially falsifying / refuting scenarios (cf. asymmetry thesis)

- Maybe theory should be: "Higher financial incentives increase performance in simple tasks **in X kind of conditions**"

THREATS TO EXTERNAL VALIDITY CRITERIA TO PAY ATTENTION TO (BERGENHOLTZ 2025A: P.4)

• Interaction of selection and treatment

- Representative participants?
- Generalizes to other people?



• Interaction of setting and treatment

- What activity is taking place, in what setting?
- Might be particular and not generalize to other activities/settings



• Interaction of history and treatment

- Society changes, studies from 1990s still valid?



• Reactive effects of experimental arrangements

- Hawthorne effect, only when observed these results occur?



BUSINESS ADMINISTRATION EXAMPLE

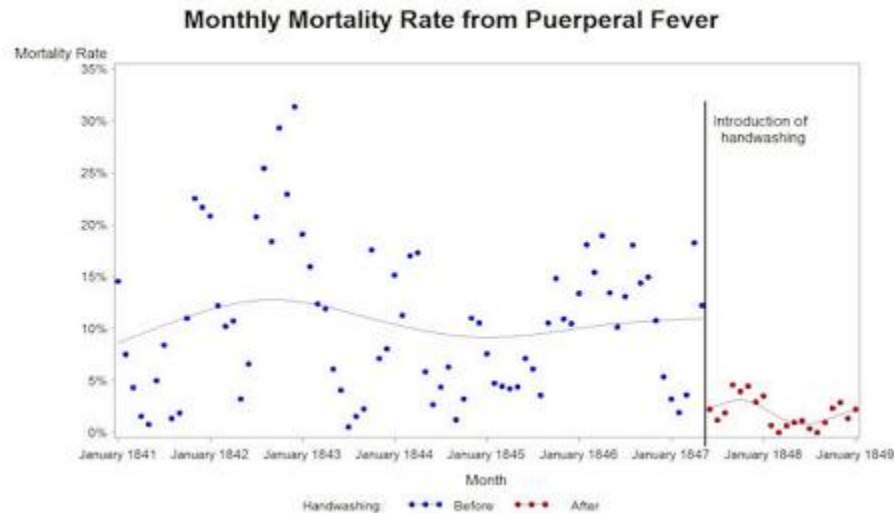
Verification approach: A marketing researcher hypothesizes that "social media presence increases customer engagement." The researcher collects data from 50 companies with active social media and finds that 42 show increased engagement metrics. The researcher concludes the theory is verified.

Problem (per Popper): When trying to confirm, risk of bias (here selection bias).

Falsification approach: Rather than find if social media presence increases engagement – see if reduction in social media presence reduces engagement.

- Test with companies that reduced social media presence
- Control for other variables
- Specify conditions under which the theory might not hold
- Accept results as tentative, not absolute truth

WHAT EXPLAINS RESISTANCE TO NEW CLAIMS: BAD METHOD OR NORMAL SCIENCE? - HANDWASHING



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Analysis of petition signers (O'Neill 2014):

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Why wasn't a better explanation accepted – quicker?

Popper:

- The medical community should have conducted objective tests to falsify hypotheses
- A critical, rational approach could have resolved it

Kuhn

- Paradigms explains resistance to germ theory/handwashing
- Not just objective evidence that leads to scientific change
- Survival of the fittest explanation...

WHAT EXPLAINS RESISTANCE TO NEW CLAIMS: BAD METHOD OR NORMAL SCIENCE? - SOCIAL MEDIA RESEARCH



Kuhn (normal science + institutions)

- Normal science”: work inside a shared paradigm (measures, questions, dependent variables etc.)
- When a big study finds weak/no effect → treated as an “anomaly” (measurement, model, sample?)
- Institutions reinforce the paradigm (journals/reviewers/reward small changes)
- Shift to new paradigm only when anomalies accumulate + a rival paradigm is established

Popper (severe tests + detachment)

- Formulate a bold, specific hypothesis (incl. direction + plausible effect size)
- Seek severe tests: preregistration, multi-site replication, challenge the idea
- If evidence contradicts hypothesis: report the null, revise/abandon theory

“Kuhn explains why fields can be slow to move; Popper is a prescription for how they **should** try to move faster.”

OK, WE SHOULD DO BOLD PROPOSITIONS AND AIM TO REFUTE AND FALSIFY - WHAT ABOUT SCIENCE AND TRUTH?



Karl Popper on Science & Absolute Truth (1974)

<https://www.youtube.com/watch?v=li0ciaqJ0m0>

POPPER: WE CAN ONLY APPROXIMATE TRUTH

— for conceit over our knowledge, or for smugness. Those among our theories which turn out to be highly resistant to criticism, and which appear to us at a certain moment of time to be better approximations to truth than other known theories, may be described, together with the reports of their tests, as 'the science' of that time. Since none of them can be positively justified, it is essentially their critical and progressive character -the fact that we can *argue* about their claim to solve our problems better than their competitors--which constitutes the rationality of science.

Theories are our own inventions, our own ideas; they are not forced upon us, but are our self-made instruments of thought: this has been clearly seen by the idealist. But some of these theories of ours can clash with reality; and when they do, we know that there is a reality; that there is something to remind us of the fact that our ideas may be mistaken. And this is why the realist is right.

Conjectures and Refutations 1962, p. 117

Post-positivism → lesson 9 (Guba)



NOTE: IF THE THEORY IS REFUTED, IT IS NOT UNSCIENTIFIC – AND PROBABLY FALSE

It is important to note that a theory does not cease to be scientific once falsified (just like an unscientific theory does not become scientific because it is true). It ceases to be valid but continues being scientific (if it was not scientific to begin with, it would not have been falsifiable at all). The only exception are cases where proponents of a refuted theory

ABH: p. 49

KUHN & POPPER: DIFFERENCES AND SIMILARITIES

Similarities:

- reject naive positivism, no direct access to reality as it is
- scientific theories are provisional, not final
- theoretical / paradigmatic frameworks shape observations

Differences

Aspect	Popper	Kuhn
Primary focus	Methodology - how science should be done	History - how science has actually developed
Truth	Science approaches truth through eliminating false theories	"Truth" is relative to paradigm; across paradigms "survival of the fittest"
Key challenge	Designing truly falsifiable theories	Overcoming resistance to paradigm change
Methodology	Deductive testing aimed at falsification	Paradigm-guided methods in normal science

2021 SPRING EXAM QUESTION (4-5 MIN)

Question 6: "A famous scientist recently claimed that science provides insight into objective reality. Please rely on the curriculum to provide counterarguments to this claim."

- How can Kuhn & Popper contribute to answer the following question? (note: additional relevant texts are presented later in the course)
- Please consider, in some detail, how you would structure an answer to the question

OUTLINE OF ANSWER TO 2021 Q6

Kuhn

- Focuses on psychological & sociological factors
- We see a processed / already interpreted version of the world (depending on education, expectations etc.)
- Therefore no theory-independent observations of the world -> all observations/data embedded in paradigms
- Does this mean relativism and anything goes?
- No: Evolutionary 'survival of the fittest' perspective on improvement of theories – not closer to reality

Popper

- Focuses on nature of induction & deduction -> what science can (not) do
- Problem of induction – next data point might prove us wrong
- Conjectures and Refutations: Create theories/hypotheses and aim to falsify – not to verify/prove
- We can always think of a different way to deductively test theory
- We can approximate truth, but will always 'only' be "science of the time"

Both agree

- No theory-neutral access to the world in itself / objective reality
- No objective truths
- Later theories often improve upon and replace earlier ones

AGENDA: SECOND HALF OF LESSON

Why business administration research is different

Nelson 2016

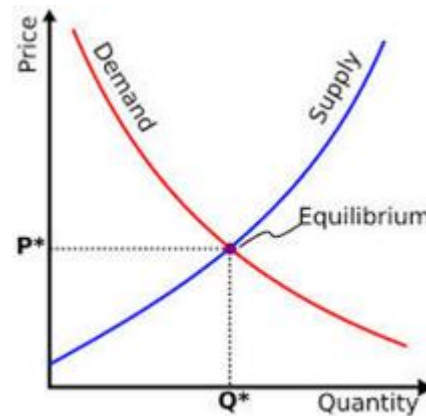
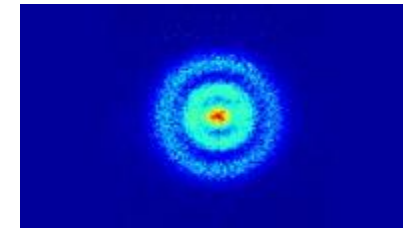
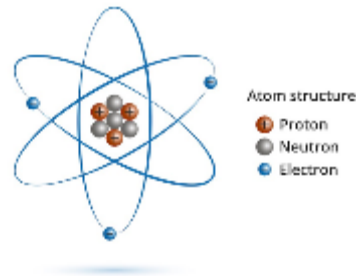
ii) How is a social science such as business administration different from (e.g.) physics and why are these differences important?

- **Earlier, related perspectives**
 - Watts 2011: Similar focus & questions (how is social science particular, compared to physics)
 - Kuhn & Popper: Focus on limits of science – in general
- **Nelson 2016:**
 - Focuses on particular challenges for business administration (e.g. vs. physics)
 - How the nature of the phenomena studied, limits what theories can do

**Following presentation of Nelson 2016 also available
in this 2021 video: <https://youtu.be/t-yLe1LlbA>**

WHAT ARE WE MEASURING, AND HOW CERTAIN CAN WE BE ABOUT OUR CONCLUSIONS?

Different approaches to these questions in different fields



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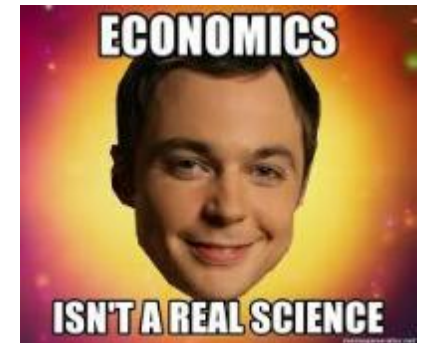
NELSON 2016: WHAT KIND OF SCIENCE IS ECONOMICS / BUSINESS ADMINISTRATION

Is business administration / economics even a science?

Overview of Nelson's claims

- We should not model social science on physics – social sciences are different (see e.g. p. 1697)
- Some other (natural) sciences are more similar to social sciences, than physics (see e.g. p. 1698-1699)
- Identifies 3 key features that distinguish business/economics from physics

In other words, how can we talk about business administration as a science?
What world are we theorizing about in different sciences?



?

THE CLAIM AND ARGUMENT, BRIEFLY PUT

Should the social sciences follow the standards and approach of physics?

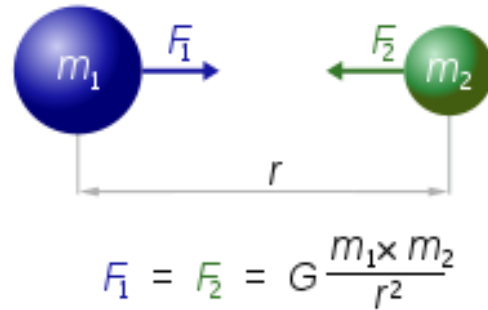
1. Introduction

In a nutshell, the argument of this essay is this. For centuries Newtonian (and later post Newtonian) physics has been regarded by both scholars who analyze science, and laypersons impressed with the accomplishments of the scientific enterprise, as the model of what a science should be. It has been the hope, and the expectation, that if sufficient resources and talent were put into the sciences concerned with other phenomena – in particular the life sciences and the behavioral and social sciences – the kind of deep and broad and precise knowledge that had been attained in the physical sciences could be attained there too, and with that knowledge the ability to design technologies, ways of doing things more generally, that would be as effective in these arenas as the technologies made possible by physics. But this is an illusion. The subject matter of the other sciences is different, and the differences matter in terms both of the kind of knowledge and power one can expect from research in these fields, and the

BUSINESS: A SCIENCE? I

WHY DOES SOCIAL SCIENCE NOT LEAD TO NEWTONIAN LAWS, LIKE IN PHYSICS?

Newtonian physics: Homogeneous phenomena and stable causal influences

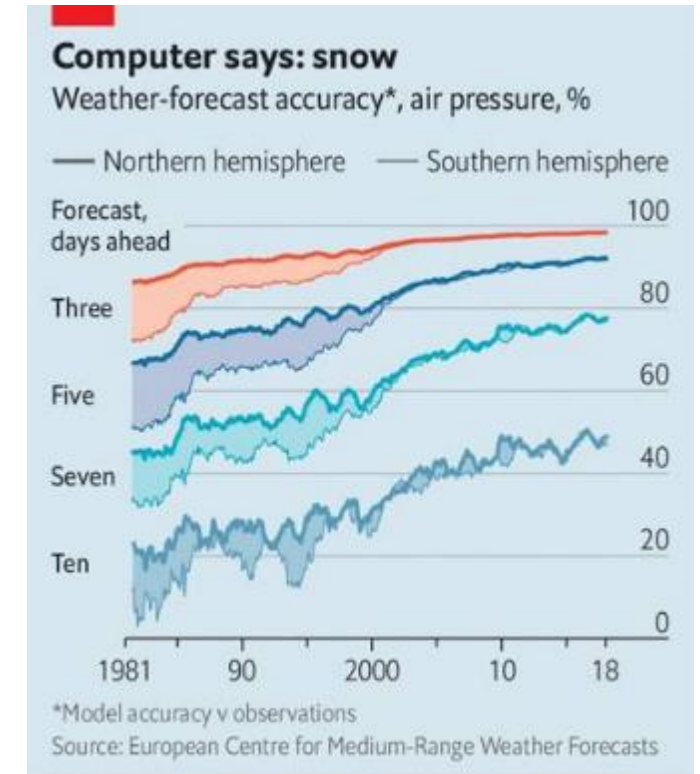


“Thus while Newton’s calculations treat only the masses of a particular planet and the sun, the effect of other masses and potential influences like galactic dust on the planet’s orbit turns out to be very small” (Nelson 2016: p. 1694)

BUSINESS: A SCIENCE? II

EVEN SEVERAL NATURAL SCIENCES AREN'T LIKE CLASSICAL PHYSICS

Meteorology (e.g. storms) $\neq$ astrophysics, Newtonian laws



The Economist

Nevertheless, meteorology is a pretty predictive science.

“Every storm is different in some often significant way ... Measures of physical characteristics of storms, like wind velocity, provide important indicators of how destructive that storm is likely to be. However, the measurements here are not precise, and in their nature cannot be, and the relevant causal connections are somewhat variable” (Nelson 2016: p. 1699)

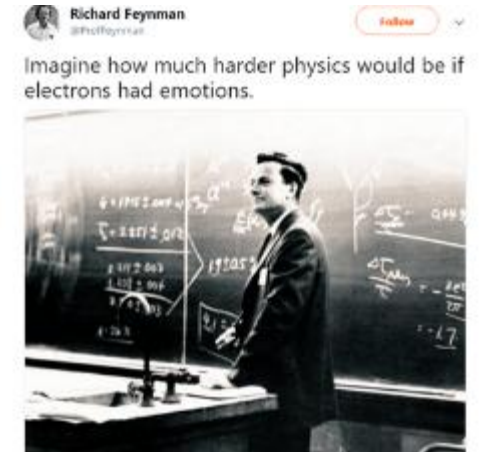
BUSINESS: A SCIENCE? III

THE WORLD(S) WE STUDY

1) “First, while the subject matter they study falls into different classes, within each class there is considerable **heterogeneity**, which cannot be reduced to a small number of well defined variables, given the questions being asked of that subject matter. And in many cases the classes themselves do not have sharply defined **boundaries**” (Nelson 2016: p. 1697)

2) “Second, the forces and conditions that **influence** the subjects of study in ways that the science seeks to understand are **numerous**, highly **variable**, and often cannot be **separated** sharply one from another. Indeed the circumstances associated with any particular set of observations may need to be understood as, in some sense, unique.” (p. 1697)

3) ...the subject matter they study often **changes over time**...” (p. 1698) – see also lesson 9

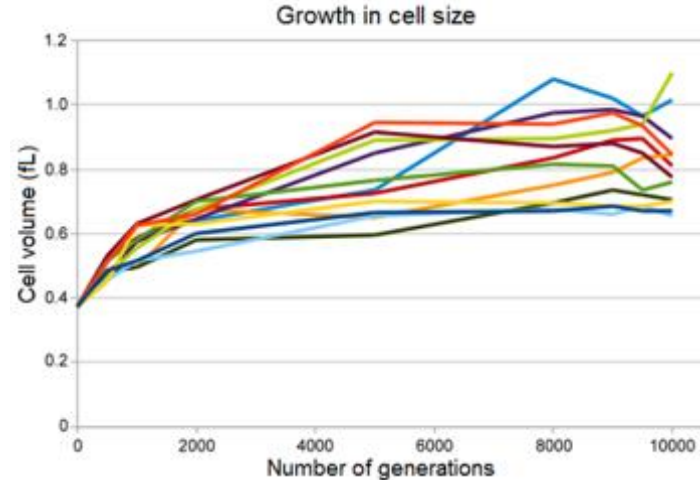


The world limits what science we can do (p. 1693)

Kuhn focused not on the world, but human/social factors
Popper focused on how nature of induction/deduction limits

HOMOGENEITY OF ATOMS OR CELLS VS. HUMAN

(NELSON POINT #1+2)



https://en.wikipedia.org/wiki/E._coli_long-term_evolution_experiment

In contrast, business administration →
"A major challenge is the heterogeneity of entrepreneurial phenomena. Different people start different types of new economic activities for different reasons; through different processes; at different pace; with different resourcebases, and in different industry, geographic, cultural, and economic environments."

<https://eprints.qut.edu.au/221162/1/95492.pdf>

Imagine letting 12 populations of humans develop separately for 50.000 generations...

EXAMPLES OF DIFFERENCES IN MEASURES - THE DIFFERENCES MATTER

3. Unemployment and GDP

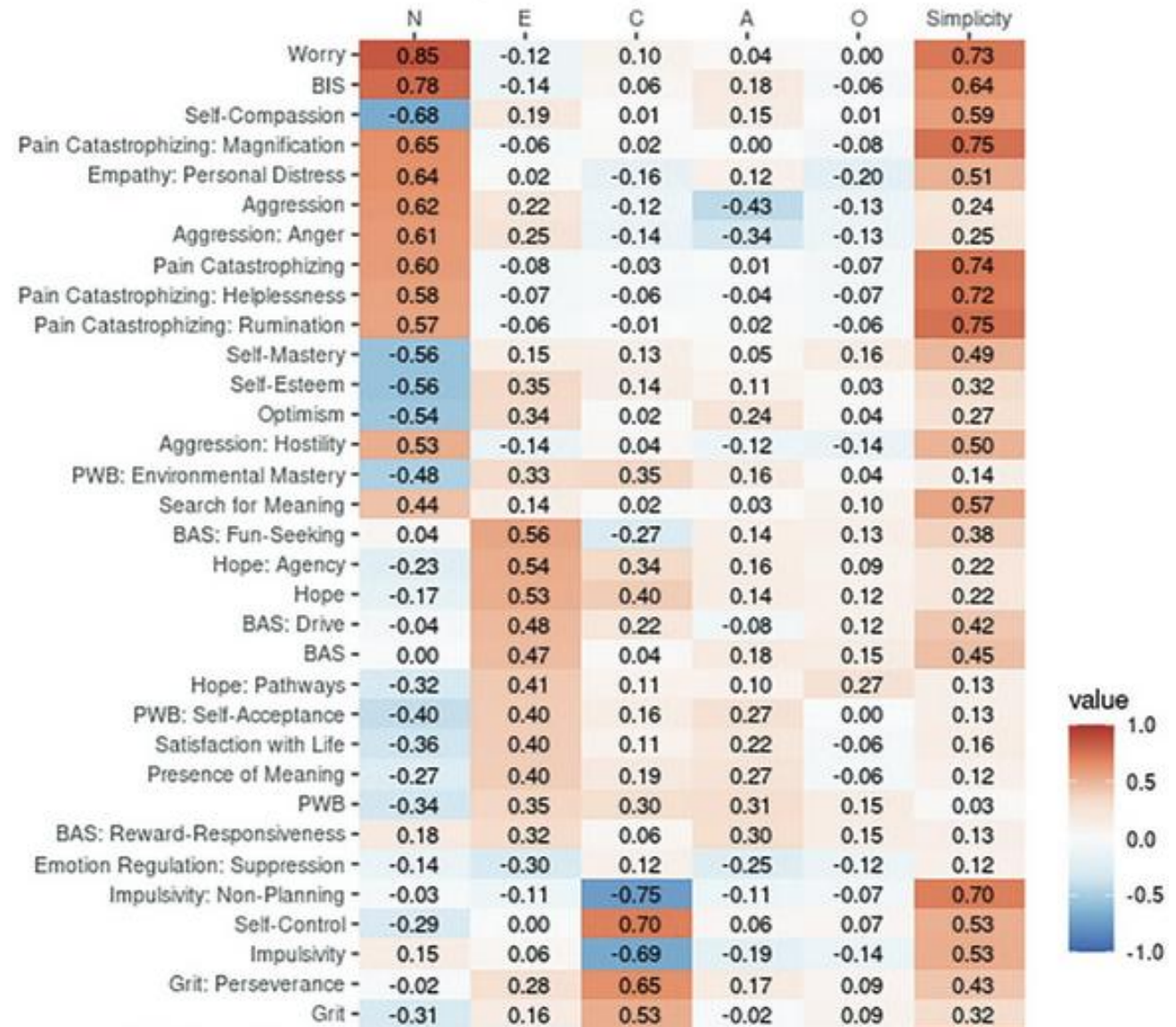
CPI measurement is not an exception. Measuring any economic indicator is a complicated business and raises conceptual, factual and normative/evaluative issues. Take unemployment as an example. The International Labour Organization (ILO) defines someone as unemployed when they are out of work, currently looking for work, and available for work. This definition is used around the world but quite vague and therefore subject to different interpretations. Using US standards, for example, Canadian unemployment is about 1 percent lower than Canada's official rate says it is (Flaherty 2012: 35). The definition says '*someone* who is looking for work...'. This does not apply to teenagers under

http://jeiss.org/Philosophy_of_Economics/PhilEcon_Manuscript_Master.pdf

BIG 5 VS OTHER PERSONALITY CONCEPTS: NO SHARP BOUNDARIES -> FUZZY OVERLAP

“I want to suggest that, where the underlying conception of the phenomenon being studied has **blurry edges, a constructed number** meant to measure an aspect of it may take on a definitional quality....”
(Nelson 2016: p. 1695)

Stand-Alone Scales' Locations in the Big Five in Study 2

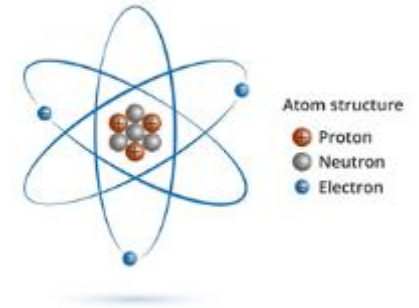


HETEROGENEOUS VARIABLES INFLUENCED BY NUMEROUS OTHER VARIABLES (NELSON POINT #1+2)

Extended Data Table 1 | Effect of professional identity on cheating in bank employees

Dependent variable:	(a)	(b)
	1 if successful outcome	
Professional identity	0.065** (0.031)	0.060** (0.030)
Age	-0.000 (0.002)	0.000 (0.003)
Male	-0.001 (0.034)	0.006 (0.033)
University degree	0.006 (0.031)	0.016 (0.034)
Foreign nationality	0.062 (0.046)	0.050 (0.049)
Relative income	-0.005 (0.012)	-0.006 (0.012)
Core business unit		0.086*** (0.033)
Years in industry		-0.000 (0.003)
Observations	1280	1280
Sample	Bank employees	Bank employees

Probit estimates. Reported results are marginal effects calculated at the median levels of the covariates. Robust standard errors corrected for clustering on the individual level are in parentheses. **a.** The decision to report a successful outcome is regressed on a dummy for the professional identity condition and individual characteristics ($n = 128$). **b.** The second model also includes job-related variables ($n = 128$). Significance levels: * $P < 0.10$, ** $P < 0.05$, *** $P < 0.01$ (two-sided Wald tests).

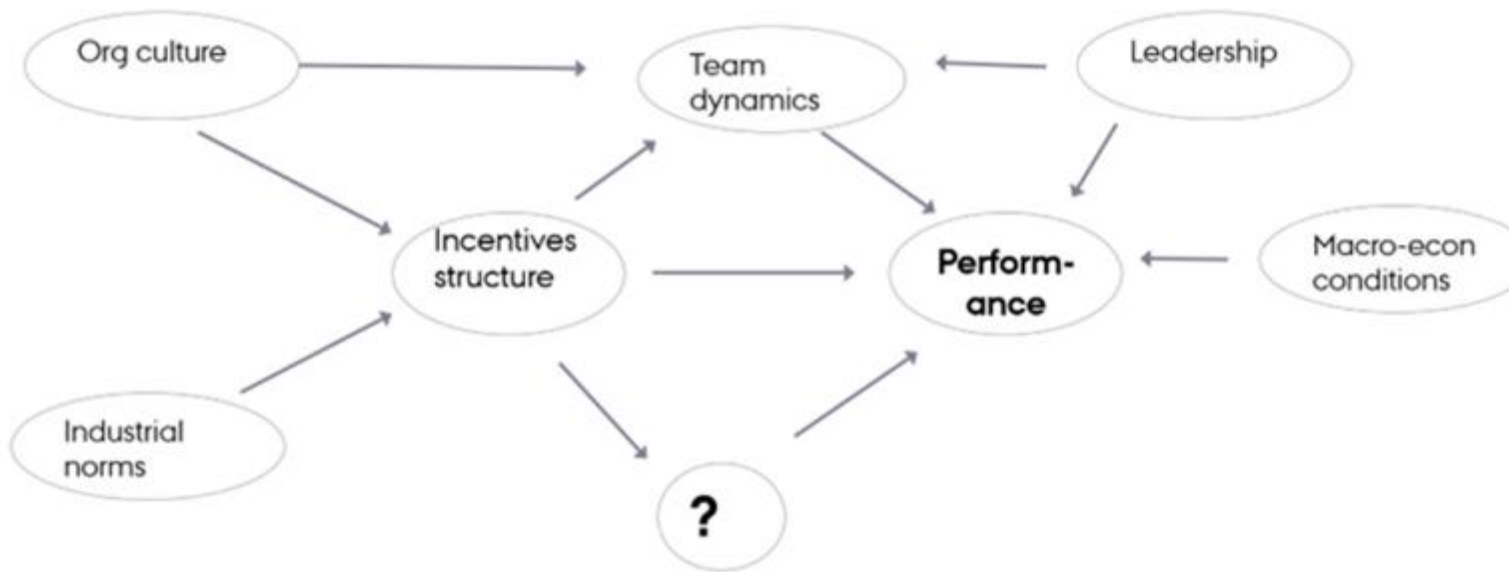


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HETEROGENEOUS VARIABLES INFLUENCED BY NUMEROUS OTHER VARIABLES (NELSON POINT #1+2)



$$F_1 = F_2 = G \frac{m_1 \times m_2}{r^2}$$



HOW MANY VARIABLES MIGHT IMPACT ORGANIZATIONAL PERFORMANCE? (NELSON POINT #2)

Market demand
Economic conditions
Industry competition
Company culture
Management effectiveness

Employee morale
Organizational structure
Innovation capacity
Financial health
Brand reputation

Customer satisfaction
Supply chain efficiency
Technological adoption
Regulatory environment
Globalization effects

Pricing strategies
Marketing campaigns
Operational processes
Legal issues
Strategic partnerships

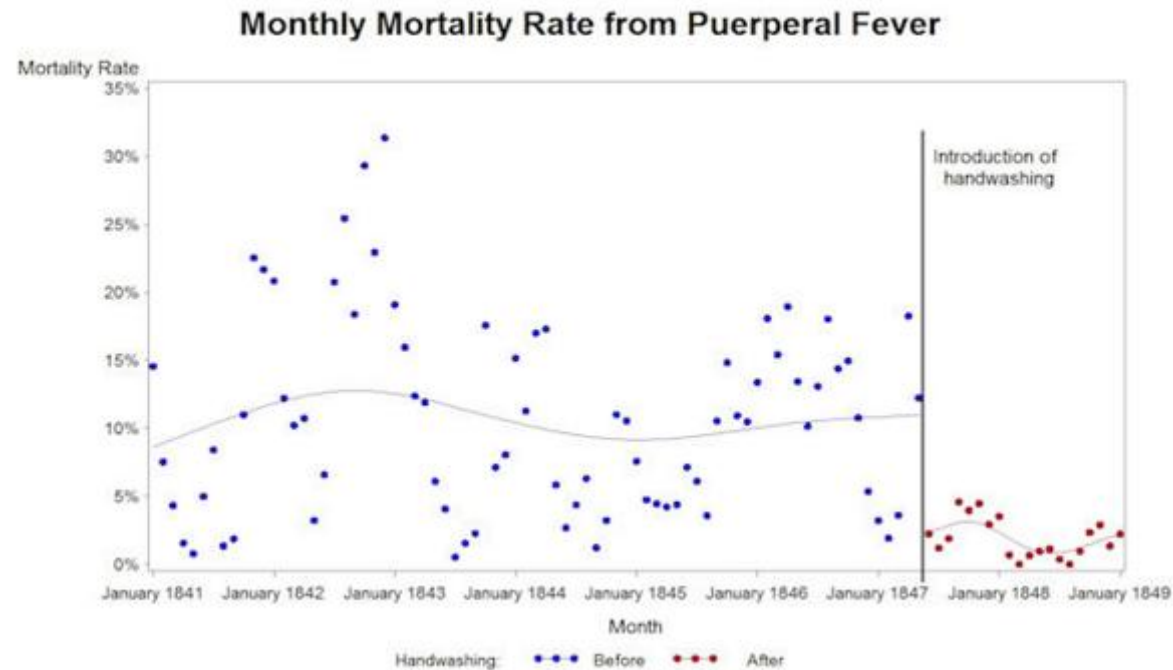
Capital availability
Talent recruitment and retention
Product quality
Service delivery
Corporate governance

Risk management
Sustainability practices
Stakeholder relationships
Digital transformation progress
Change management capabilities

And the chatbot could have generated many more

COMPARE WITH MEDICINE (SEMMELWEIS) EXAMPLE

MORE DISTINCT VARIABLES: EASIER TO MEASURE IF SOMEONE IS DEAD (OR NOT)



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PEOPLE / ORGANISATIONS ARE HETEROGENEOUS + CAN CHANGE BEHAVIOR OVER TIME (NELSON POINT #3)



Kieran Healy
@kjhealy

Following

Think of social science as a small bit like hurricane forecasting, except the hurricane is able to watch the news.



- **Original study:** opposite-race referees called more fouls on Black players (1991-2002 data)
- **Media awareness:** Racial bias widely discussed in NBA / media in 2007 when published
- **Follow-up study:** No racial bias in 2007-2010 data

PEOPLE / ORGANISATIONS CAN CHANGE BEHAVIOR OVER TIME (NELSON POINT #3)

In this issue, Lori's paper with Ram Ranganathan and Anindya Ghosh explored how **taken-for-granted determinants of alliance formations might vary across industries and timeframes**. Some typical predictors – such as product-market similarity between two potential alliance partners – generated robust positive results across the industries and timeframes analyzed. **Yet others – such as previous alliances between a pair of firms – yielded dramatically different results in the chemicals industry: Previous alliances were positively associated with future alliances in the 1980s, but negatively associated with them in the 1990s.**

https://lorirosenkopf.substack.com/p/from-quasi-replication-to-generalization?r=1v5hrv&utm_campaign=post&utm_medium=web

NELSON ON LIMITS OF EXPERIMENTS (P. 1699)



How homogeneous and predictable is the world?

Are the different schools in the same treatment **really** identical?

In recent years the cutting edge of the design of empirical research on the efficacy of educational practices has been to assign students randomly to the treatment being studied, and to compare how they do with the performance of a control group assigned to other things.²¹ For many years this has been the preferred way of studying the effects of different treatments in medicine. This research strategy provides a way of assessing whether a particular generally broadly defined way of doing something is on average efficacious in comparison with another practice or treatment or arrangement. However, when applied to practice in education there almost always are differences in exactly what a nominal practice is between schools in a school system, or even within schools between classrooms. And almost always students vary in their responses to different treatments. Further, there are good reasons to doubt that another empirical study done in ways that are as similar as possible would yield results just like the first study.



NELSON ON (DANGER OF) DREAM OF HOMO ECONOMICUS AND ASSUMING ALL SCIENCES ARE LIKE PHYSICS (P. 1700)

An exaggerated belief in the range of phenomena where scientific research can help us see the variables of interest in terms of a few, well defined numbers and molded by clear sharp laws has led many scientists and policy makers to push hard for scientific methods that assume that the phenomena they want illuminated can be seen this way. I do not want to denigrate the value of numbers where they do measure or indicate important aspects of the phenomena of interest, or the value of statistical analysis of factors influencing variables measured this way, or of mathematical models that explore the implications of causal relationships that are believed to exist. But if the phenomena of interest are of the kind I have been describing, this form of analysis can tell us, at best, only part of the story we want to know about, and even that part can only be interpreted sensibly in the context of a broader, more qualitative account. The problem I have tried to highlight is that the belief I have been discussing leads us to value too highly “scientific findings” that purport to be exact and clean, where it is dubious that the phenomena being studied in fact can be described that simply or behave that regularly, and at the same time to undervalue the kind of illumination that science can provide that is of a more qualitative form.

We risk misguided science and science policy/investments

If we expect precise science, why fund science that is not precise?

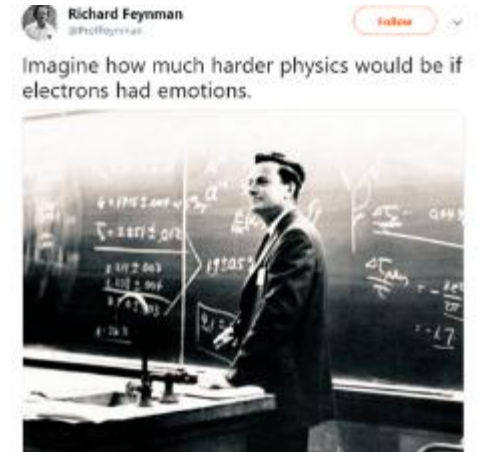
BUSINESS: A SCIENCE? III

THE WORLD(S) WE STUDY

1) “First, while the subject matter they study falls into different classes, within each class there is considerable **heterogeneity**, which cannot be reduced to a small number of well defined variables, given the questions being asked of that subject matter. And in many cases the classes themselves do not have sharply defined **boundaries**.” (Nelson 2016: p. 1697)

2) “Second, the forces and conditions that **influence** the subjects of study in ways that the science seeks to understand are **numerous**, highly **variable**, and often cannot be **separated** sharply one from another. Indeed the circumstances associated with any particular set of observations may need to be understood as, in some sense, unique.” (p. 1697)

3) ...the subject matter they study often **changes over time**...” (p. 1698) – see also lesson 9



The world limits what science we can do (p. 1693)

Kuhn focused not on the world, but human/social factors
Popper focused on how nature of induction/deduction limits

HOW CAN THESE TOPICS APPEAR IN AN EXAM SITUATION?

(REFLECT ON THIS EXAM QUESTION, 2-3 MIN)

2018: Question 6 (23%)

Will business administration researchers ever be able to come up with clear, universal theories about all individual and business behavior; e.g., is it possible to create a clear, universal theory about how all firms should be innovative to succeed? (just like the theory of gravity informs us about the motion of all physical entities) Please provide reasons for your answer and base it on the curriculum and course material.

Additional angles on this question will be presented later in the course

**Extracts from an
actual exam
answer**

When discussing the possibility of creating clear, universal theories, ontological and epistemological considerations come into play. If one believes it is possible to clearly define an independent reality and that the theories which are proposed represent this reality perfectly, you have a positivistic point of view. The theory of gravity can clearly predict the motion of objects in a true way. However, theories within business administration such as motivational theories cannot perfectly predict how motivated an employee will be given certain factors. This contradicts with this positivistic view point and relates more to a post-positivistic or constructivist viewpoint. From a constructivist viewpoint, reality only exists within a context, thus we, as people, construct reality in our minds. Theories are constructed and do not represent the world as it really is. Instead, they are based on subjective knowledge. Therefore, they cannot predict behavior in a perfect way.

Nelson (2016) points out the difference between sciences very well. Physics is often seen as the model for other sciences, but we cannot compare sciences as such since they are very different. As he puts it *"First, while the subject matter they study falls into different classes, within each class there is considerable heterogeneity, which cannot be reduced to a small number of well defined variables, given the questions being asked of that subject matter"* (Nelson, 2016). Physics deal with subjects which can be clearly defined and the variables affecting them can be broken down to a small number and be well-defined. This is not the case for most other sciences such as social sciences in which business administration works. Social sciences deal with very heterogeneous subjects that often can be hard to define or can be defined in many different ways. For example, innovation – how would one clearly define that? This can be hard and often people will not agree on how to define it.

Nelson continues: *“Second, the forces and conditions that influence the subjects of study in ways that the science seeks to understand are numerous, highly variable and often cannot be separated sharply one from another”* (Nelson, 2016). The variables which influence the subjects are numerous and they vary a lot. They can be very hard to define, some might not even be known, they can be hard to separate and they are often interrelated. If we were to find all variables which can influence innovation it would take a very long time and different people would claim that different variables are relevant. Furthermore, these variables would have an effect on each other when being changed leading to another effect on innovation. This makes it very difficult to isolate one variable’s effect in order to establish a causal relationship. This is also why experiments can be hard to set up within the social world. How should we isolate all other effects?

These characteristics about subjects and variables make social sciences very different from physics. This difference has a great impact on the findings and conclusions which are made by social science researchers. As Nelson puts it: *“For a science to be able to find and characterize stable and relatively precise laws and to make reliable quantitative predictions, the subject matter it studies needs to be relatively homogeneous, with variation a matter of a few dimensions and capable of clean specification, with similar homogeneity and separability of the different forces and conditions that affect the behaviors of that subject matter”* (Nelson, 2016). This is not the case in social sciences nor business administration. Because of the large number of variables, their blurry boundaries and their interrelated nature it is hard to come up with a perfect theory that perfectly predicts future behavior in social sciences.

‘YOUR’ MEASURES AND PHENOMENA, E.G. IN YOUR BACHELOR PROJECT (REFLECT, 1-2 MIN)

How easy is it to measure / make sense of what you are interested in?

1) “First, while the subject matter they study falls into different classes, within each class there is considerable **heterogeneity** [=heterogeneitet, uens], which cannot be reduced to a small number of well defined variables, given the questions being asked of that subject matter. And in many cases the classes themselves do not have sharply defined **boundaries** [=grænser]. “ (Nelson 2016: p. 1697)

2) “Second, the forces and conditions that **influence** the subjects of study in ways that the science seeks to understand are **numerous**, highly **variable**, and often cannot be **separated** [=adskilt] sharply one from another. Indeed the circumstances associated with any particular set of observations may need to be understood as, in some sense, unique.” (p. 1697)

CONSIDER THIS: IS IT A GOOD THING IF 100% OF SCIENTISTS AGREE ON SOMETHING? (REFLECT 1-2 MIN)

BUSINESS: A SCIENCE? III THE WORLD(S) WE STUDY

1) "First, while the subject matter they study falls into different classes, within each class there is considerable **heterogeneity** [=heterogenität, vers], which cannot be reduced to a small number of well defined variables, given the questions being asked of that subject matter. And in many cases the classes themselves do not have sharply defined **boundaries** [=grenzen]." [Nelson 2016: p. 1697]

2) "Second, the forces and conditions that **influence** the subjects of study in ways that the science seeks to understand are **numerous**, highly **variable**, and often cannot be **separated** [=abtrennen] sharply one from another. Indeed the circumstances associated with any particular set of observations may need to be understood as, in some sense, unique." (p. 1697)

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- **Kuhn:** consensus is a feature - enables normal science.
- **Popper:** consensus is a risk – might protect theories from criticism
- **Nelson:** In business administration consensus is harder to reach – due to heterogeneity + change

SUMMING UP: IMPLICATIONS OF NELSON'S CHARACTERISTIC OF BUSINESS ADMINISTRATION / ECONOMICS

- Physics' quantitative "law-focus" should not be the goal of business administration
- We don't investigate homogeneous, unchanging phenomena that can be easily isolated
- When phenomena and theoretical relations are heterogeneous, we have to acknowledge limitations & be transparent in our categorization of our data/variables
- Both quantitative and qualitative work important – selection of method should be based on specific problem field
- The more homogeneous the field, the easier it is to quantify average relations

SUMMING UP I: KUHN, POPPER & NELSON

Aspect	Popper	Kuhn	Nelson
Primary focus	Methodology - how science should be done	History - how science has actually developed	How subject matter constrains scientific methods
Truth	Science approaches truth through eliminating false theories	"Truth" is relative to paradigm; across paradigms "survival of the fittest"	Context-dependent, limited by heterogeneity of phenomena
Key challenge	Designing truly falsifiable theories	Overcoming resistance to paradigm change	Accepting heterogeneity without expecting physics-like precision
Methodology	Deductive testing aimed at falsification	Paradigm-guided methods in normal science	Multi-method approach, due to heterogeneous, complex phenomena

SUMMING UP II: WHY READ KUHN, POPPER AND NELSON?

They have shaped scientists' (and societies') ideas about what science is

- How rational and objective science is
- What objectivity / kind of truth can we expect
- How are some sciences different than others
- Established counterpoints to positivism & a naive conception of science



Overall: Important to acknowledge weaknesses and strengths of science

- Not a flaw that scientists disagree. Don't hope for 100% consensus
- It's ok that theories we thought were true, turn out to be false